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Task 1 – Por olio Website Design

You are building a personal por olio website to showcase your work.

Requirements:

- Create sec ons for About Me, Projects, and Contact.
- Use AI to:
 - o Suggest color pale es and typography.
 - o Create a responsive layout with Grid/Flexbox.
 - o Add smooth scrolling naviga on.

PROMPT:

"Create a single-page personal portfolio website with 'About Me', 'Projects', and 'Contact' sections.

CODE:

```
18 portfolio.html > <html> <body>
19   <!DOCTYPE html>
20   <html lang="en">
21   <head>
22     <meta charset="UTF-8">
23     <meta name="viewport" content="width=device-width, initial-scale=1.0">
24     <title>John Doe | Personal Portfolio</title>
25
26     <!-- Google Fonts -->
27     <link rel="preconnect" href="https://fonts.googleapis.com">
28     <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
29     <link href="https://fonts.googleapis.com/css2?family=Montserrat:wght@400;500;600&display=swap" rel="stylesheet">
30
31     <!-- Custom CSS -->
32     <link rel="stylesheet" href="style.css">
33   </head>
34   <body>
35
36     <!-- Header & Navigation -->
37     <header class="main-header">
38       <nav class="navbar">
39         <a href="#about" class="nav-link">About</a>
40         <a href="#projects" class="nav-link">Projects</a>
41         <a href="#contact" class="nav-link">Contact</a>
42       </nav>
43     </header>
44
45     <!-- Main Content Area -->
46     <main>
47       <div class="hero">
48         <h1>Hi, I'm John Doe</h1>
49         <p>A Creative Professional</p>
50       </div>
51
52       <div class="about">
53         <h2>About Me</h2>
54         <p>I am a creative professional with a passion for design and technology. I have worked on various projects, from web development to digital marketing. I am currently looking for new opportunities to grow and learn.</p>
55       </div>
56
57       <div class="projects">
58         <h2>Projects</h2>
59         <ul>
60           <li>Project 1: A full-stack web application for managing a team's tasks and projects.</li>
61           <li>Project 2: A responsive website for a local business, featuring a clean and modern design.</li>
62           <li>Project 3: A digital marketing campaign for a new product launch, including social media posts and email newsletters.</li>
63         </ul>
64       </div>
65
66       <div class="contact">
67         <h2>Contact</h2>
68         <p>Get in touch with me for any inquiries or collaborations. I am always open to new ideas and challenges.</p>
69         <form>
70           <input type="text" value="Name" />
71           <input type="email" value="Email" />
72           <input type="text" value="Subject" />
73           <input type="text" value="Message" />
74           <input type="button" value="Send" />
75         </form>
76       </div>
77     </main>
78
79     <!-- Footer -->
80     <footer>
81       <p>© 2024 John Doe. All rights reserved. <a href="#">Privacy Policy</a> <a href="#">Terms of Service</a></p>
82     </footer>
83   </body>
84 </html>
```

```

28 </header>
29
30 <main>
31   <!-- About Me Section -->
32   <section id="about" class="section-container">
33     <h1>Hi, I'm John Doe</h1>
34     <p class="subtitle">A passionate software developer creating modern and responsive web applications.</p>
35     <p>I specialize in front-end and back-end development, with a strong focus on user experience and code quality. I love turning
36   </section>
37
38   <!-- Projects Section -->
39   <section id="projects" class="section-container">
40     <h2>My Projects</h2>
41     <div class="projects-grid">
42       <!-- Project Card 1 -->
43       <div class="project-card">
44         <h3>Project One</h3>
45         <p>A web application for managing tasks and boosting productivity. Built with React and Node.js.</p>
46         <a href="#" class="btn">View Project</a>
47       </div>
48       <!-- Project Card 2 -->
49       <div class="project-card">
50         <h3>Project Two</h3>
51         <p>An e-commerce platform with a custom CMS, developed using Python (Django) and PostgreSQL.</p>
52         <a href="#" class="btn">View Project</a>
53       </div>
54       <!-- Project Card 3 -->
55       <div class="project-card">
56         <h3>Project Three</h3>
57         <p>A data visualization dashboard for analyzing market trends. Created with D3.js and a RESTful API.</p>
58         <a href="#" class="btn">View Project</a>
59       </div>
60     </div>
61   </section>
62
63   <!-- Contact Section -->
64   <section id="contact" class="section-container">
65     <h2>Get In Touch</h2>
66     <p>Have a question or want to work together? Feel free to reach out!</p>
67     <a href="mailto:email@example.com" class="btn btn-primary">Say Hello</a>
68   </section>
69 </main>
70
71 <script src="script.js"></script>
72 </body>
73 </html>

```

```

# style.css > {} @media (max-width: 768px)
1 /* --- 1. AI-Suggested Color Palette & Typography --- */
2 root {
3   --bg-color: #f0f0f0;
4   --primary-text-color: #000000;
5   --secondary-text-color: #666666;
6   --accent-color: #008080;
7   --card-bg-color: #e0e0e0;
8   --border-color: #444;
9
10  --font-heading: 'Montserrat', sans-serif;
11  --font-body: 'Lato', sans-serif;
12 }
13
14 /* --- Basic Reset & Body Styles --- */
15 * {
16   margin: 0;
17   padding: 0;
18   box-sizing: border-box;
19 }
20
21 html {
22   /* This enables smooth scrolling via CSS, but we'll use JS for broader browser support */
23   scroll-behavior: smooth;
24 }
25
26 body {
27   font-family: var(--font-body);
28   background-color: var(--bg-color);
29   color: var(--primary-text-color);
30   line-height: 1.6;
31 }

```

```

34 .main-header {
35   background-color: rgba(26, 26, 26, 0.9);
36   backdrop-filter: blur(10px);
37   padding: 1rem 5%;
38   position: fixed;
39   top: 0;
40   left: 0;
41   width: 100%;
42   z-index: 1000;
43   border-bottom: 1px solid var(--border-color);
44 }
45
46 .navbar {
47   display: flex;
48   justify-content: space-between;
49   align-items: center;
50   max-width: 1200px;
51   margin: 0 auto;
52 }
53
54 .nav-logo {
55   font-family: var(--font-heading);
56   font-size: 1.5rem;
57   color: var(--accent-color);
58   text-decoration: none;
59 }
60
61 .nav-menu {
62   display: flex;
63   list-style: none;
64   gap: 2rem;
65 }

```

```

67 .nav-link {
68   color: var(--primary-text-color);
69   text-decoration: none;
70   font-weight: 700;
71   transition: color 0.3s ease;
72 }
73
74 .nav-link:hover {
75   color: var(--accent-color);
76 }
77
78 /* --- General Section & Content Styling --- */
79 .section-container {
80   padding: 6rem 5% 2rem;
81   max-width: 1200px;
82   margin: 0 auto;
83   min-height: 80vh;
84   border-bottom: 1px solid var(--border-color);
85 }
86
87 h1, h2 {
88   font-family: var(--font-heading);
89   color: var(--primary-text-color);
90   margin-bottom: 1rem;
91 }
92
93 h1 { font-size: 3rem; }
94 h2 { font-size: 2.5rem; color: var(--accent-color); }
95 .subtitle { font-size: 1.2rem; color: var(--secondary-text-color); margin-bottom: 2rem; }
96
97 /* --- Projects Section (Flexbox) --- */
98 .projects-grid {
99   display: flex;

```

```

100     flex-wrap: wrap;
101     gap: 1.5rem;
102     margin-top: 2rem;
103 }
104
105 .project-card {
106     background-color: var(--card-bg-color);
107     border: 1px solid var(--border-color);
108     border-radius: 8px;
109     padding: 1.5rem;
110     flex: 1 1 300px; /* flex-grow, flex-shrink, flex-basis */
111     transition: transform 0.3s ease, box-shadow 0.3s ease;
112 }
113
114 .project-card:hover {
115     transform: translateY(-5px);
116     box-shadow: 0 8px 20px rgba(0, 170, 255, 0.1);
117 }
118
119 .project-card h1 {
120     font-family: var(--font-heading);
121     color: var(--accent-color);
122     margin-bottom: 0.5rem;
123 }
124
125 /* --- Button Styling --- */
126 .btn {
127     display: inline-block;
128     margin-top: 1rem;
129     padding: 0.75rem 1.5rem;
130     background-color: var(--accent-color);
131     color: var(--bg-color);

```

```

130     background-color: var(--accent-color);
131     color: var(--bg-color);
132     text-decoration: none;
133     font-weight: 700;
134     border-radius: 5px;
135     transition: background-color 0.3s ease;
136 }
137
138 .btn:hover {
139     background-color: #0088cc;
140 }
141
142 /* --- Responsive Design --- */
143 @media (max-width: 768px) {
144     h1 { font-size: 2.5rem; }
145     h2 { font-size: 2rem; }
146     .nav-menu { display: none; } /* Simple hiding for mobile, could be replaced with a hamburger menu */
147 }

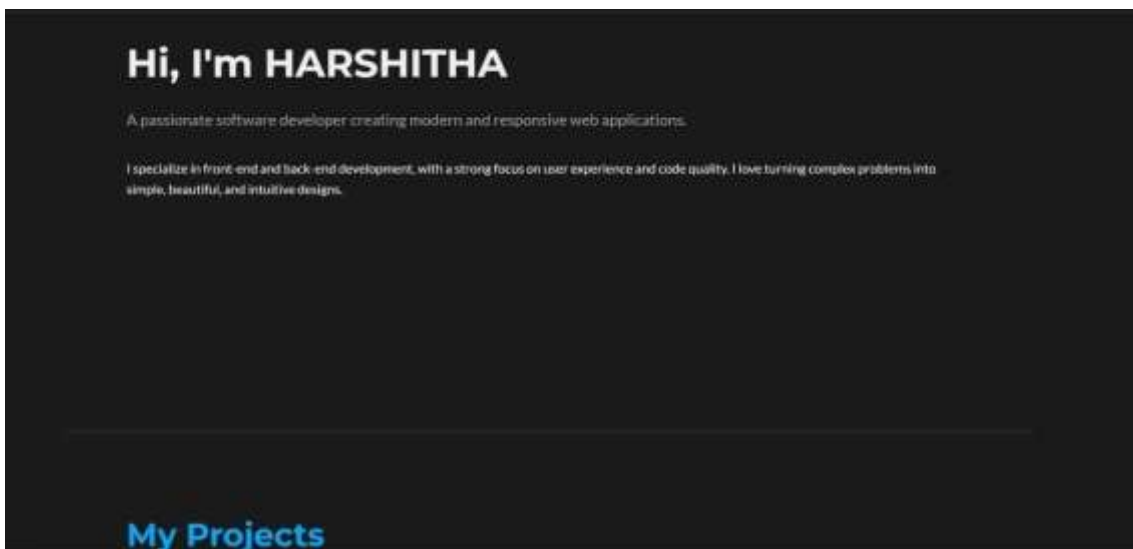
```

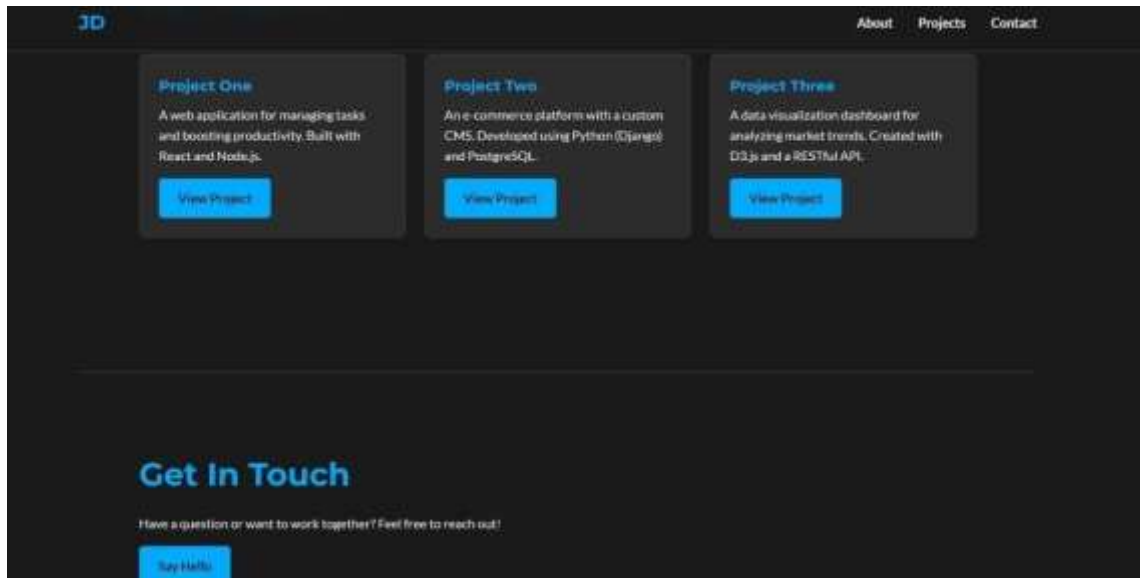
```

18 script.js > ...
1 document.addEventListener('DOMContentLoaded', function() {
2   // Select all navigation links
3   const navLinks = document.querySelectorAll('.nav-link, .nav-logo');
4
5   navLinks.forEach(link => {
6     link.addEventListener('click', function(e) {
7       // Prevent the default anchor link behavior
8       e.preventDefault();
9
10      // Get the target section's ID from the href attribute
11      const targetId = this.getAttribute('href');
12      const targetSection = document.querySelector(targetId);
13
14      // Smoothly scroll to the target section
15      targetSection.scrollIntoView({ behavior: 'smooth', block: 'start' });
16    });
17  });
18 });

```

OUTPUT:





OBSERVATION:

- **Color and Typography:** The AI-suggested palette (`:root` variables in CSS) creates a modern, high-contrast dark mode that is easy on the eyes. The combination of 'Montserrat' for headings and 'Lato' for body text provides a clean, professional look that is highly readable.
- **Responsive Layout:**
 - **Flexbox** is used in `.navbar` to space out the logo and the menu links, and in `.projects-grid` to create a flexible, wrapping container for the project cards. The `flex: 1 1 300px` property on `.project-card` is powerful: it allows cards to grow and shrink but ensures they have a base width of 300px, causing them to wrap naturally on smaller screens.
 - The `@media` query demonstrates a basic approach to mobile responsiveness by hiding the navigation menu on smaller screens. A more advanced implementation would add a "hamburger" menu icon to toggle the navigation.
- **Smooth Scrolling:** The JavaScript solution is more robust than the CSS-only `scroll-behavior: smooth;`. It intercepts the click event on each navigation link, prevents the default "jump," and then uses the `scrollIntoView({ behavior: 'smooth' })` method to perform a clean, animated scroll. This provides a better user experience and has wider browser compatibility for this specific type of interaction.

Task 2 – Online Store Product Page

Design a product display page for an online store.

Requirements:

- Display product image, title, price, and "Add to Cart" button.
- Use AI to:
 - Style with BEM methodology.
 - Make layout responsive.
 - Add hover effects and "Add to Cart" alert.

PROMPT:

"Create the HTML, CSS, and JavaScript for a single product display page for an online store.

Page Content & Structure:

1. The page must display a product image, a product title, a price, and an "Add to Cart" button.

CODE:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Product Page</title>
7   <link rel="stylesheet" href="product-style.css">
8 </head>
9 <body>
10
11   <div class="product-card">
12     <!-- Product Image Section -->
13     <div class="product-card_image-container">
14       
15     </div>
16
17     <!-- Product Details Section -->
18     <div class="product-card_details">
19       <h3 class="product-card_title">AuraSound Wireless Headphones</h3>
20       <p class="product-card_price">$149.99</p>
21       <p class="product-card_description">Experience immersive audio with these sleek, noise-cancelling wireless headphones. Perfect for travel, work, or relaxation.
22       <button class="product-card_button product-card_button-primary" id="addToCartBtn">Add to Cart</button>
23     </div>
24   </div>
25
26   <script src="product-script.js"></script>
27
28 </body>
29 </html>
```



```

# product-styles.sass ---
1  /* --- General Body Styles --- */
2  body {
3    font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Helvetica, Arial, sans-serif;
4    background-color: #fafafa;
5    display: flex;
6    justify-content: center;
7    align-items: center;
8    min-height: 100vh;
9    margin: 0;
10   padding: 1rem;
11 }
12
13 /* --- BEM Block: product-card --- */
14 .product-card {
15   display: flex;
16   background-color: #ffffff;
17   border-radius: 12px;
18   box-shadow: 0 10px 30px rgba(0, 0, 0, 0.1);
19   overflow: hidden;
20   max-width: 900px;
21   width: 100%;
22 }
23
24 /* --- BEM Element: product-card_image-container --- */
25 .product-card_image-container {
26   flex: 1 1 50%; /* Grow, Shrink, Basis */
27 }
28
29 /* --- BEM Element: product-card_image --- */
30 .product-card_image {
31   width: 100%;
32   height: 100%;
33   object-fit: cover; /* Ensures the image covers the area without distortion */
34 }

```

```

35
36 /* --- BEM Element: product-card_details --- */
37 .product-card_details {
38   flex: 1 1 50%;
39   padding: 2rem;
40   display: flex;
41   flex-direction: column;
42   justify-content: center;
43 }
44
45 /* --- BEM Element: product-card_title --- */
46 .product-card_title {
47   font-size: 2rem;
48   font-weight: 700;
49   margin: 0 0 0.5rem;
50   color: #1a1a1a;
51 }
52
53 /* --- BEM Element: product-card_price --- */
54 .product-card_price {
55   font-size: 1.75rem;
56   font-weight: 500;
57   color: #007bff;
58   margin: 0 0 1rem;
59 }
60
61 /* --- BEM Element: product-card_description --- */
62 .product-card_description {
63   font-size: 1rem;
64   line-height: 1.5;
65   color: #808080;

```

```

66     margin-bottom: 2rem;
67   }
68
69   /* --- BEM Element: product-card_button --- */
70   .product-card_button {
71     border: none;
72     padding: 1rem 2rem;
73     font-size: 1rem;
74     font-weight: 600;
75     border-radius: 8px;
76     cursor: pointer;
77     transition: background-color 0.3s ease, transform 0.2s ease;
78   }
79
80   /* --- BEM Modifier: product-card_button--primary --- */
81   .product-card_button--primary {
82     background-color: #007bff;
83     color: #ffffff;
84   }
85
86   .product-card_button--primary:hover {
87     background-color: #0056b3;
88     transform: translateY(-2px); /* Subtle lift effect on hover */
89   }
90
91   /* --- Responsive Layout --- */
92   @media (max-width: 768px) {
93     .product-card {
94       flex-direction: column; /* Stack image and details vertically */
95     }
96   }

```

```

JS product-script.js > ...
1  document.addEventListener('DOMContentLoaded', function() {
2    // Find the 'Add to Cart' button by its ID
3    const addToCartButton = document.getElementById('addToCartBtn');
4
5    // Add a click event listener to the button
6    addToCartButton.addEventListener('click', function() {
7      // Show a simple alert when the button is clicked
8      alert('AuraSound Wireless Headphones have been added to your cart!');
9    });
10 });

```

OUTPUT:



OBSERVATION:

Observations

- **BEM Methodology:** The CSS is highly organized and readable due to the BEM naming convention. `product-card` is the "Block," `product-card_title` is an "Element," and `product-card_button--primary` is a "Modifier." This structure makes the code self-documenting and easy to maintain.
- **Responsive Design:** The use of Flexbox (`display: flex`) on the `.product-card` is key. The simple media query at the bottom of the CSS file changes `flex-direction: column;` on screens narrower than 768px, which elegantly handles the responsive requirement without extra HTML.
- **User Feedback:** The hover effect on the button (`transform: translateY(-2px);`) provides a satisfying "lift" that signals interactivity. The JavaScript `alert()` gives immediate and clear confirmation of the user's action, fulfilling the requirement for feedback.

This solution provides a robust and professional foundation for a product page that is both visually appealing and highly functional.

Task 3 – Event Registration Form

Build an event registration form for a conference.

Requirements:

- Collect name, email, phone number, and session selection.
- Use AI to:
 - o Add form validation with JavaScript.
 - o Make the form accessible with labels and ARIA.
 - o Style with a professional look.

PROMPT:

"Create the HTML, CSS, and JavaScript for an event registration form for a tech conference.

Form Fields:

1. Full Name (text input)
2. Email Address (email input)
3. Phone Number (tel input)
4. Session Selection (a dropdown/select menu with at least 3 options and a disabled default option).

CODE:

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Event Registration</title>
7      <link rel="stylesheet" href="form-style.css">
8  </head>
9  <body>
10
11      <div class="form-container">
12          <h1 class="form-title">Conference Registration</h1>
13          <form id="registrationForm" novalidate>
14              <!-- Name Field -->
15              <div class="form-group">
16                  <label for="fullName">Full Name</label>
17                  <input type="text" id="fullName" name="fullName" aria-required="true">
18                  <div class="error-message" id="nameError"></div>
19              </div>
20
21              <!-- Email Field -->
22              <div class="form-group">
23                  <label for="email">Email Address</label>
24                  <input type="email" id="email" name="email" aria-required="true">
25                  <div class="error-message" id="emailError"></div>
26              </div>
27
28              <!-- Phone Field -->
29              <div class="form-group">
30                  <label for="phone">Phone Number</label>
31                  <input type="tel" id="phone" name="phone" aria-required="true">
32                  <div class="error-message" id="phoneError"></div>
33              </div>
```

```

29     <div class="form-group">
30         <label for="phone">Phone Number</label>
31         <input type="tel" id="phone" name="phone" aria-required="true">
32         <div class="error-message" id="phoneError"></div>
33     </div>
34
35     <!-- Session Selection Field -->
36     <div class="form-group">
37         <label for="session">Choose a Session</label>
38         <select id="session" name="session" aria-required="true">
39             <option value="" disabled selected>Select a session...</option>
40             <option value="ai-ml">AI & Machine Learning</option>
41             <option value="web-dev">Modern Web Development</option>
42             <option value="cloud">Cloud Native Architectures</option>
43         </select>
44         <div class="error-message" id="sessionError"></div>
45     </div>
46
47     <button type="submit" class="submit-btn">Register</button>
48 </form>
49 <div id="successMessage" class="success-message"></div>
50 </div>
51
52 <script src="form-script.js"></script>
53
54 </body>
55 </html>

```

```

# form-style.css --
1  /* --- General Body & Font Styles --- */
2  body {
3      font-family: system-ui, -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, Oxygen, Ubuntu, Cantarell, 'Open Sans',
4      background-color: #eef2f7;
5      display: flex;
6      justify-content: center;
7      align-items: center;
8      min-height: 100vh;
9      padding: 1rem;
10 }
11
12 /* --- Form Container Styling --- */
13 .form-container {
14     background-color: #ffffff;
15     padding: 2.5rem;
16     border-radius: 10px;
17     box-shadow: 0 4px 20px rgba(0, 0, 0, 0.08);
18     width: 100%;
19     max-width: 500px;
20 }
21
22 .form-title {
23     text-align: center;
24     margin-bottom: 2rem;
25     color: #333;
26     font-weight: 600;
27 }
28
29 /* --- Form Group & Label Styling --- */
30 .form-group {
31     margin-bottom: 1.5rem;
32 }

```

```

35     display: block;
36     margin-bottom: 0.5rem;
37     font-weight: 500;
38     color: #555;
39 }
40
41 /* --- Input & Select Styling --- */
42 input[type="text"],
43 input[type="email"],
44 input[type="tel"],
45 select {
46     width: 100%;
47     padding: 0.75rem;
48     border: 1px solid #ccc;
49     border-radius: 5px;
50     font-size: 1rem;
51     transition: border-color 0.3s ease, box-shadow 0.3s ease;
52 }
53
54 input:focus,
55 select:focus {
56     outline: none;
57     border-color: #007bff;
58     box-shadow: 0 0 0.3px rgba(0, 123, 255, 0.2);
59 }
60
61 /* --- Validation & Error Styling --- */
62 .error-message {
63     color: #dc3545;
64     font-size: 0.875rem;
65     margin-top: 0.25rem;
66     min-height: 1em; /* prevents layout shift */

```

```

69 .success-message {
70     color: #28a745;
71     text-align: center;
72     font-weight: 500;
73     margin-top: 1rem;
74 }
75
76 /* Add a class for invalid inputs */
77 .invalid {
78     border-color: #dc3545;
79 }
80
81 .invalid:focus {
82     box-shadow: 0 0 0.3px rgba(220, 53, 69, 0.2);
83 }
84
85 /* --- Button Styling --- */
86 .submit-btn {
87     width: 100%;
88     padding: 0.85rem;
89     background-color: #007bff;
90     color: #fff;
91     border: none;
92     border-radius: 5px;
93     font-size: 1.1rem;
94     font-weight: 600;
95     cursor: pointer;
96     transition: background-color 0.3s ease;
97 }
98
99 .submit-btn:hover {
100     background-color: #0056b3;

```

OUTPUT:

Conference Registration

Full Name

Email Address

Phone Number

Choose a Session

Select a session...▼

Register

OBSERVATION:

- **Accessibility:** The form is highly accessible. Every input has a corresponding `<label>` with a `for` attribute, which is crucial for screen reader users to understand what each field is for. The `aria-required="true"` attribute further informs assistive technologies that the field must be filled out.
- **JavaScript Validation:** The `novalidate` attribute on the `<form>` tag disables the browser's default validation popups, allowing our custom JavaScript to take full control. The script provides a superior user experience by:
 - Checking all fields at once on submission.
 - Displaying clear, specific error messages directly on the page.
 - Using CSS (`.invalid` class) to visually highlight problematic fields.
- **Professional Styling:** The CSS creates a modern and professional look. Key features include:
 - A soft background color for the page and a clean white card for the form.
 - Subtle `box-shadow` to make the form "float."
 - Clear visual feedback on `:focus` for inputs, improving usability.
 - A responsive design that ensures the form looks great on both desktop and mobile devices.

Task Descrip on #4 (Data – Fetch API & Render List with Loading/Error States)

- Task: Fetch JSON from an API and render items to the DOM with loading and error UI.
- Instruc ons: o Ask AI to write `fetch()` logic, create DOM nodes safely, and add skeleton/loading text.

Deliverables (For All Tasks)

1. AI-generated prompts for code and test case genera on.
2. At least 3 assert test cases for each task.
3. AI-generated ini al code and execu on screenshots.
4. Analysis of whether code passes all tests.
5. Improved final version with inline comments and explana on.

6. Compiled report (Word/PDF) with prompts, test cases, assertions, code, and output.

PROMPT:

"Create an HTML page that fetches a list of users from the JSONPlaceholder API (<https://jsonplaceholder.typicode.com/users>) and displays their names and emails.

CODE:

```
data.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>Fetch API Data</title>
7    <link rel="stylesheet" href="data-style.css">
8  </head>
9  <body>
10   <h1>User List</h1>
11   <div id="user-container" class="user-container">
12     <!-- Content will be dynamically inserted here -->
13   </div>
14   <script src="data-script.js"></script>
15 </body>
16 </html>
```

```
# data-style.css > body
1 body {
2   font-family: sans-serif;
3   background-color: #fef2f5;
4   color: #333;
5   padding: 2rem;
6 }
7
8 h1 {
9   text-align: center;
10  color: #1d2129;
11 }
12
13 .user-container {
14   max-width: 600px;
15   margin: 0 auto;
16   background-color: #fff;
17   border-radius: 8px;
18   padding: 1rem;
19   box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
20 }
21
22 /* --- Success State: User Item Styling --- */
23 .user-item {
24   padding: 1rem;
25   border-bottom: 1px solid #ddd;
26 }
27
28 .user-item:last-child {
29   border-bottom: none;
30 }
31
32 .user-name {
33   font-weight: bold;
```

```
33     font-weight: bold;
34     font-size: 1.1rem;
35     margin: 0 0 0.25rem;
36 }
37
38 .user-email {
39     color: #666;
40     font-size: 0.9rem;
41     margin: 0;
42 }
43
44 /* --- Loading State: Skeleton Screen Styling --- */
45 .skeleton {
46     padding: 1rem;
47     border-bottom: 1px solid #ddd;
48 }
49
50 .skeleton-text {
51     height: 1em;
52     background-color: #e0e0e0;
53     border-radius: 4px;
54     animation: pulse 1.5s infinite ease-in-out;
55 }
56
57 .skeleton-text.title {
58     width: 40%;
59     margin-bottom: 0.5rem;
60 }
61
62 .skeleton-text.subtitle {
63     width: 60%;
64 }
```

```
55 }
56
57 .skeleton-text.title {
58   width: 40%;
59   margin-bottom: 0.5rem;
60 }
61
62 .skeleton-text.subtitle {
63   width: 60%;
64 }
65
66 @keyframes pulse {
67   0% { background-color: ■ #e0e0e0; }
68   50% { background-color: ■ #f0f0f0; }
69   100% { background-color: ■ #e0e0e0; }
70 }
71
72 /* --- Error State Styling --- */
73 .error-message {
74   color: ■ #d93025;
75   font-weight: bold;
76   text-align: center;
77   padding: 2rem;
78 }
```

```

25 data-script.js > _
1 document.addEventListener('DOMContentLoaded', () => {
2   const userContainer = document.getElementById('user-container');
3   const API_URL = 'https://jsonplaceholder.typicode.com/users';
4
5   // --- UI Rendering Functions ---
6
7   /**
8    * Renders the initial skeleton loading state.
9    */
10  function renderLoadingState() {
11    userContainer.innerHTML = ''; // Clear previous content
12    // Create 5 skeleton placeholders
13    for (let i = 0; i < 5; i++) {
14      const skeleton = document.createElement('div');
15      skeleton.className = 'skeleton';
16      skeleton.innerHTML = `
17        <div class="skeleton-text title"></div>
18        <div class="skeleton-text subtitle"></div>
19      `;
20      userContainer.appendChild(skeleton);
21    }
22  }
23
24  /**
25   * Renders the list of users from the fetched data.
26   * @param {Array} users - An array of user objects.
27   */
28  function renderSuccessState(users) {
29    userContainer.innerHTML = ''; // Clear previous content
30    users.forEach(user => {
31      // Create elements safely to prevent XSS
32      const userItem = document.createElement('div');
33      userItem.className = 'user-item';

```

```

34
35      const nameEl = document.createElement('p');
36      nameEl.className = 'user-name';
37      nameEl.textContent = user.name; // Use textContent for safety
38
39      const emailEl = document.createElement('p');
40      emailEl.className = 'user-email';
41      emailEl.textContent = user.email; // Use textContent for safety
42
43      userItem.appendChild(nameEl);
44      userItem.appendChild(emailEl);
45      userContainer.appendChild(userItem);
46    });
47  }
48
49  /**
50   * Renders an error message.
51   * @param {string} message - The error message to display.
52   */
53  function renderErrorState(message) {
54    userContainer.innerHTML = ''; // Clear previous content
55    const errorEl = document.createElement('p');
56    errorEl.className = 'error-message';
57    errorEl.textContent = message;
58    userContainer.appendChild(errorEl);
59  }

```

```

05  //
06  async function fetchUsers() {
07      renderLoadingState();
08      // Test Case #1 Assertion: Check if loading state is rendered
09      console.assert(document.querySelector('.skeleton'), "Test 1 Failed: Skeleton loader should be visible.");
10
11      try {
12          const response = await fetch(API_URL);
13          if (!response.ok) {
14              throw new Error('Network response was not ok (Status: ${response.status})');
15          }
16          const users = await response.json();
17          renderSuccessState(users);
18
19          // Test Case #2 Assertion: Check if data is rendered correctly
20          const userItems = document.querySelectorAll('.user-item');
21          console.assert(userItems.length === users.length, "Test 2 Failed: Expected ${users.length} user items, but found");
22
23      } catch (error) {
24          console.error('Fetch error:', error);
25          renderErrorState('Failed to load users. Please try again later.');
```

OUTPUT:

User List
Leanne Graham Sincere@april.biz
Ervin Howell Thomas@melissa.tv
Clementine Bauch Nathan@jessica.net
Patricia Lebsack Juliana.O'Connor@kory.org
Chelsey Dietrich Ludie_Hettinger@jerrie.ca
Mrs. Dennis Schulist Karley_Dach@jasper.info
Kurtis Weissnat Telly_Hunter@kelly.biz
Nicholas Runolfsdottir V Shanevot@rosamond.me
Glennia Reichert Chase_McDermott@dana.io

Observe on:

- **User Experience (UX):** The skeleton screen is a significant improvement over a simple "Loading..." text. It reduces perceived wait time and prevents page layout shifts when the real content loads. The clear error message is also crucial for a good user experience.
- **Code Quality & Security:**
 - The logic is cleanly separated into functions for each state (`renderLoadingState`, `renderSuccessState`, `renderErrorState`), making the code easy to read and maintain.
 - Using `async/await` with a `try...catch` block is the modern standard for handling asynchronous operations like `fetch`.
 - **The most critical aspect is security.** The code correctly uses `document.createElement` and sets `element.textContent`. This automatically sanitizes the input, preventing malicious HTML or script tags in the API data from being executed (an XSS attack). Using `element.innerHTML = ...` would have been insecure.
- **Testing:** The `console.assert()` statements serve as simple, effective integration tests. They verify that the correct UI is being rendered in response to different API outcomes, confirming that the core logic works as expected.